

Principles of Experimental Design

MATH 213

Experiment vs Observational Study

An *observational study* is a study in which the researcher does not actively control the value of any variable, but simply observes the values as they naturally exist

An *experiment* is a study in which the researcher actively controls one or more of the explanatory variables

Principles of experimental design

1. **Controlling:** Assign treatments to cases and do one's best to control any other differences in groups.
E.g. Some patients take a sip of water with a pill. Others drink a whole glass of water. Researchers might tell everyone to drink an 8 oz glass of water to control for the effect of water consumption.
2. **Randomization:**
 - a. Randomly sample from the population (to be able to generalize to the population) and
 - b. Randomly assign subjects to treatments (to minimize confounding variables to be able to draw causal conclusions) whenever possible.

Principles of experimental design, ctd.

3. Replication: Within a study, replicate by collecting a sufficiently large sample. Or replicate the entire study.

4. Blocking: If there are variables that are known or suspected to affect the response variable, first group subjects into blocks based on these variables, and then randomize cases within each block to treatment groups.

E.g. If we are looking at the effect of a drug on heart attacks, we might block based on heart attack risk-level. This strategy ensures that each treatment group has the same number of low-risk patients and the same number of high-risk patients.

[Link to picture in textbook.](#)

Reducing bias in human experiments

- **Placebo:** fake treatment, often used as the control group for medical studies
- **Placebo effect:** occurs when subjects show improvement simply because they believe they are receiving a special treatment
- **Blinding:** when the subjects do not know whether they are in the control or treatment group
- **Double-blind:** when both the experimental units and the researchers who interact with the subjects do not know who is in the control and who is in the treatment group